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Clustering Analysis using Prototype Adjustment and DBSCAN

Task 1 - Overview

Student ID: R00277227

Last digit (a): 7

Second-last digit (b): 2

Random Seed: seed = 100 + 10×7 + 2 = 172

Dataset A - Gaussian Clusters

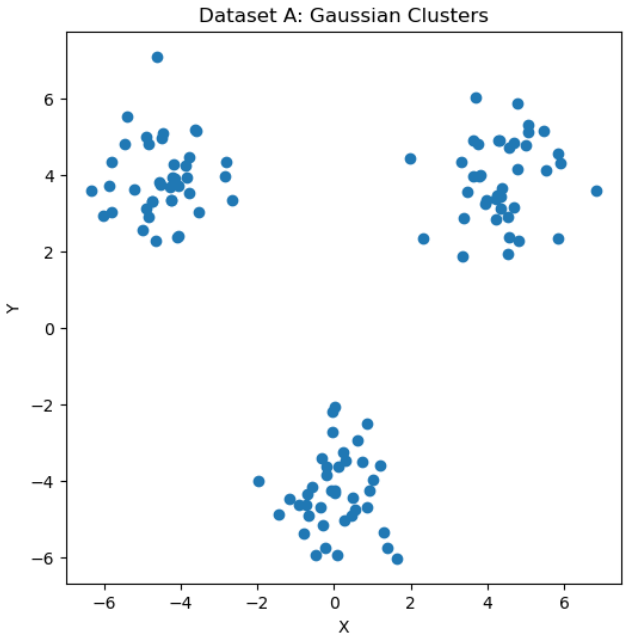
Dataset A consists of 120 points drawn from three Gaussian clusters (40 points each), each with identity covariance matrix (variance = 1 in all directions). The cluster centres derived from the student-specific parameters are:

Cluster 1 centre: (-4 - 0.1×7, 4 + 0.1×2) = (-4.7, 4.2)

Cluster 2 centre: (0, -4 - 0.1×7) = (0, -4.7)

Cluster 3 centre: (4 + 0.1×2, 4) = (4.2, 4.0)

```
Dataset A Statistics:
Mean (X, Y): -0.020435092792113953 1.1555824895377806
Std Dev (X, Y): 3.711871864961598 3.976466659778563
```

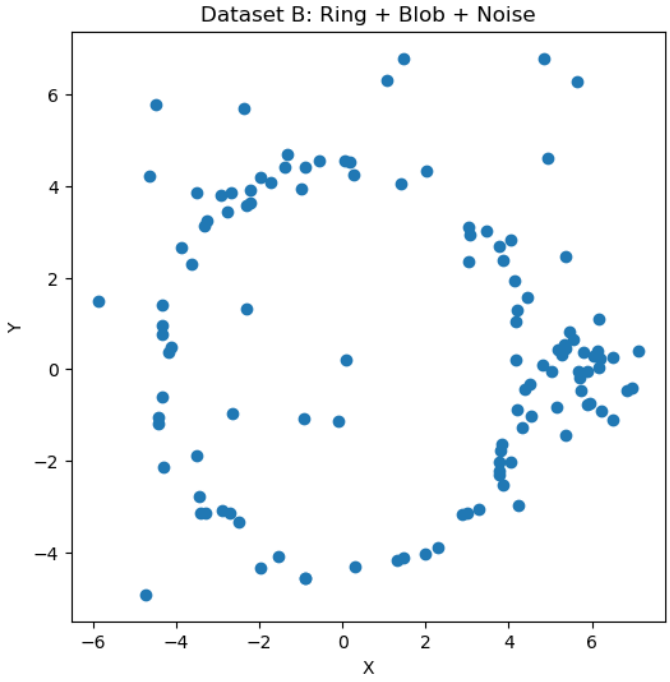


Dataset A - Raw Gaussian Clusters Scatter Plot

Dataset B - Ring + Blob + Noise

Dataset B consists of 120 points from three sources combined: 70 ring points generated using a uniform angle θ ~ U(0, 2π) and radius r = 4.5 + ε where ε ~ N(0, 0.25^2); 30 blob points centred at (5 + 0.1×7, 0) = (5.7, 0) with covariance 0.5·I; and 20 uniform noise points sampled from [−7, 7] × [−7, 7].

```
Dataset B Statistics:
Mean (X, Y): 1.4257629604251596 0.5597705964919366
Std Dev (X, Y): 3.820895176728409 2.907670475451719
```



Dataset B - Ring + Blob + Noise Scatter Plot

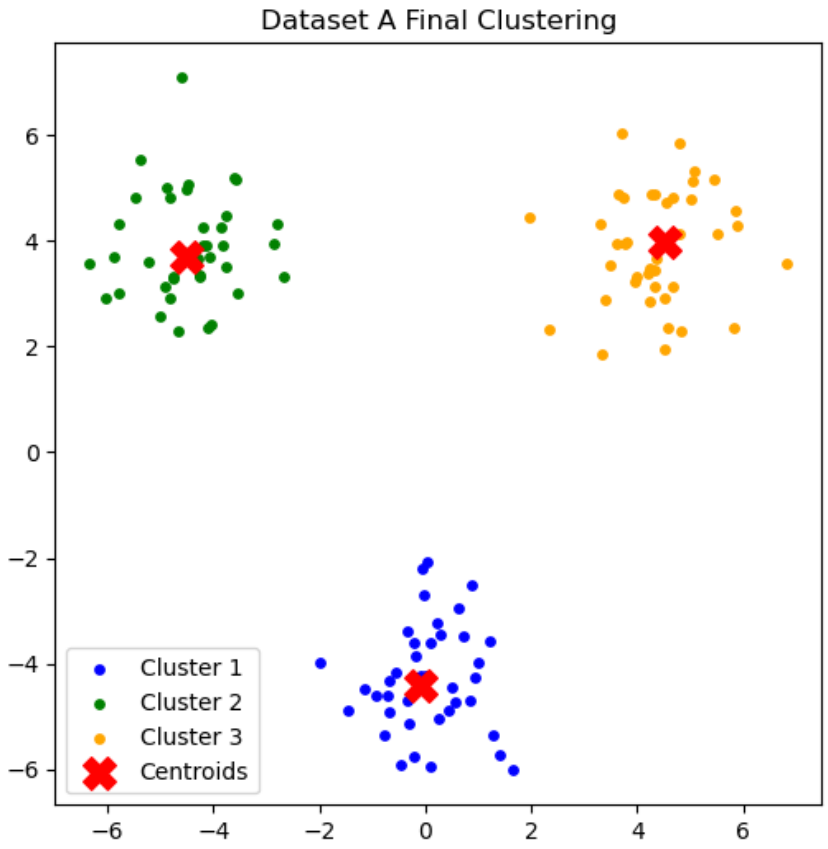
Task 2: Manual Prototype Adjustment Clustering

2.1 Algorithm Overview

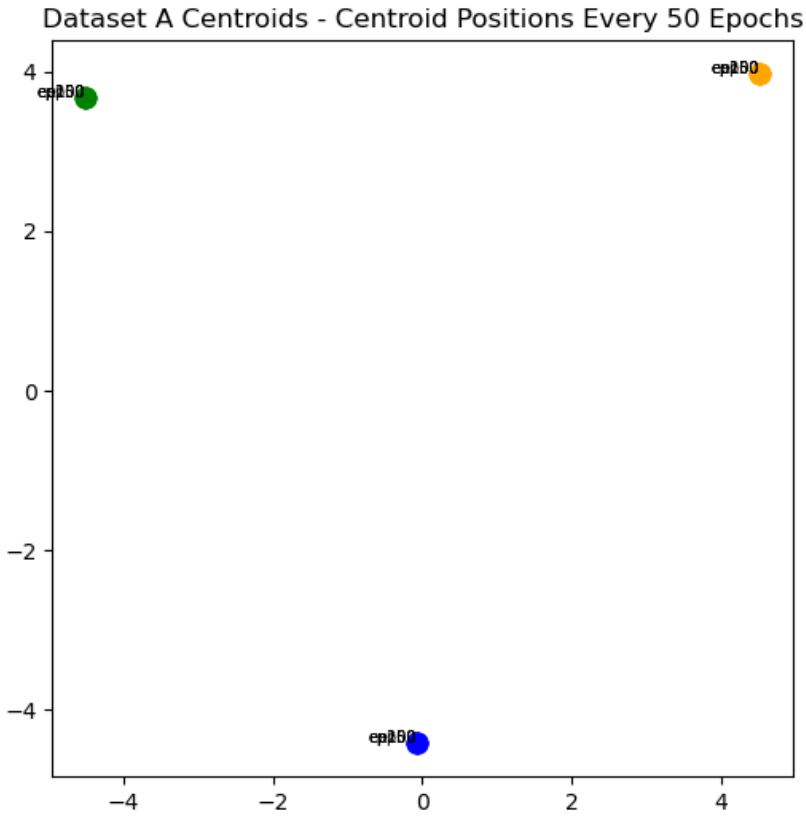
The prototype adjustment algorithm operates as an online (point-by-point) gradient-free clustering method. Centroids are initialised deterministically: the first centroid is placed at the point with minimum x + y; each subsequent centroid is placed at the point farthest from all previously selected centroids. At each epoch, every data point is assigned to its nearest centroid using Euclidean distance, and that centroid is immediately updated by a fixed step α in the direction of the point along each axis. Empty clusters are resolved by applying a small random offset drawn from U(−0.5, 0.5). The algorithm runs for 200 epochs. Sum of Squared Errors (SSE) is computed at the end of each epoch.

2.2 Dataset A Results (K = 3, α = 0.05)

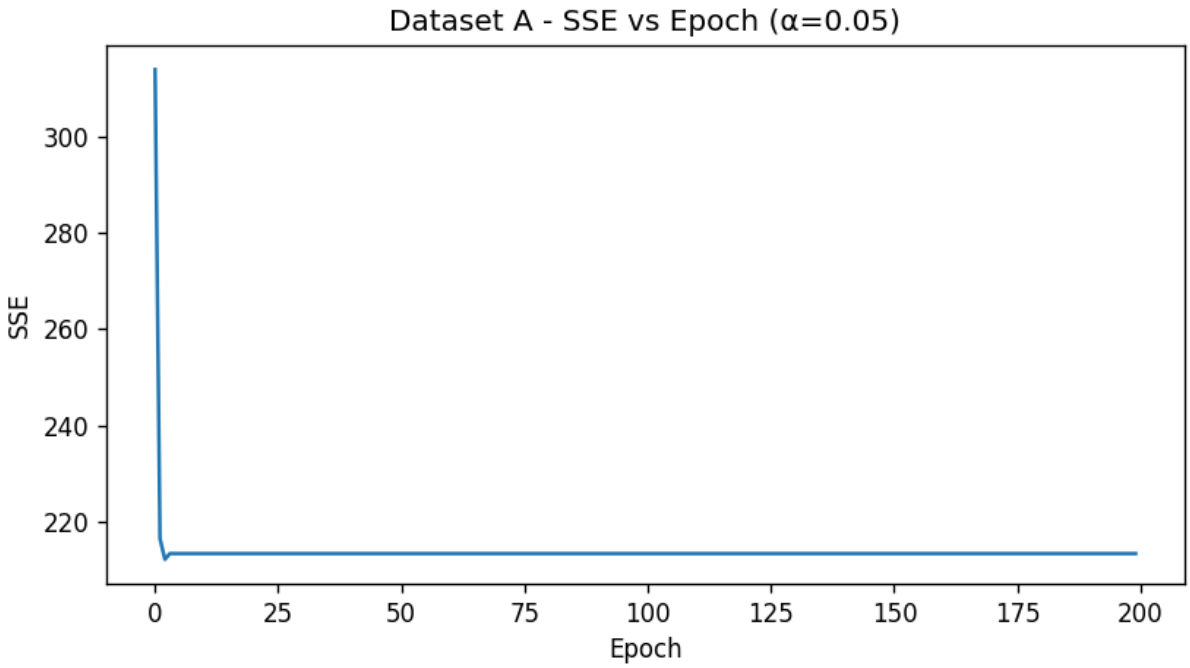
```
Final Centroids A:
 [[-0.07632561 -4.41690798]
 [-4.50753894  3.68515655]
 [ 4.5242403   3.96983369]]
```



Dataset A Final Prototype Clustering with Centroids



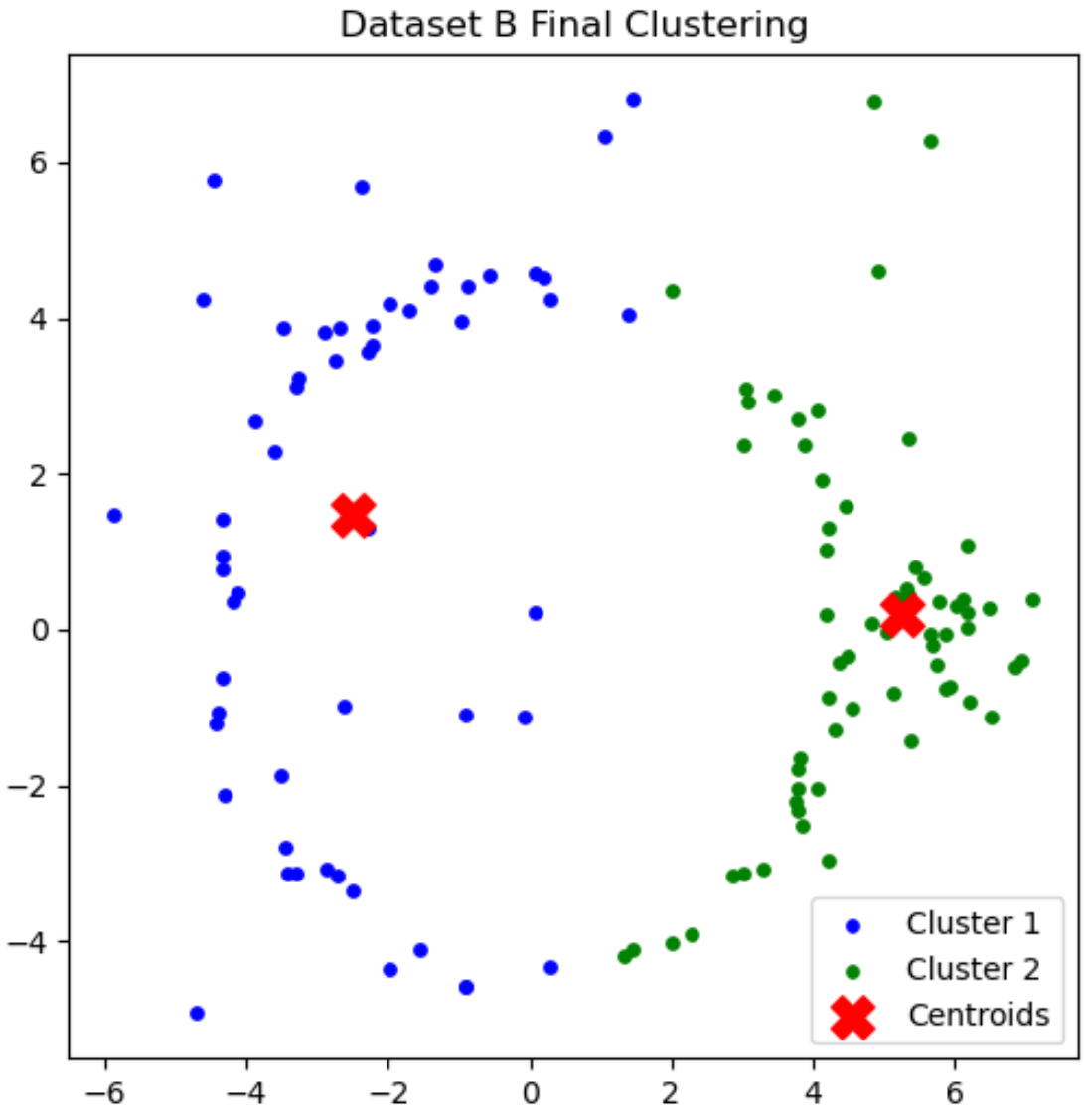
Dataset A - Centroid Positions at Epochs, 50, 100, 150, 200



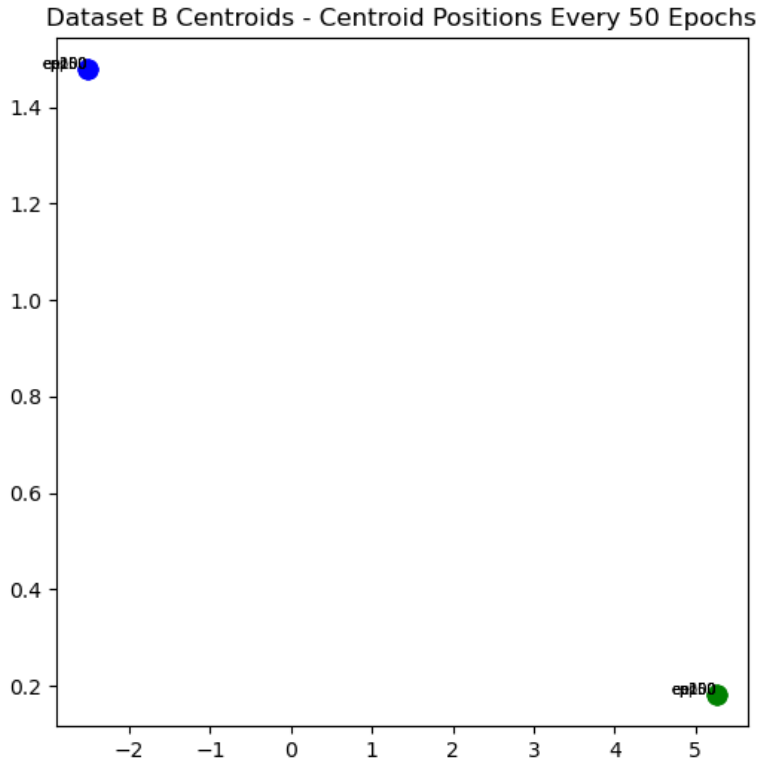
Dataset A - SSE vs Epoch (α = 0.05)

2.3 Dataset B Results (K = 2, α = 0.05)

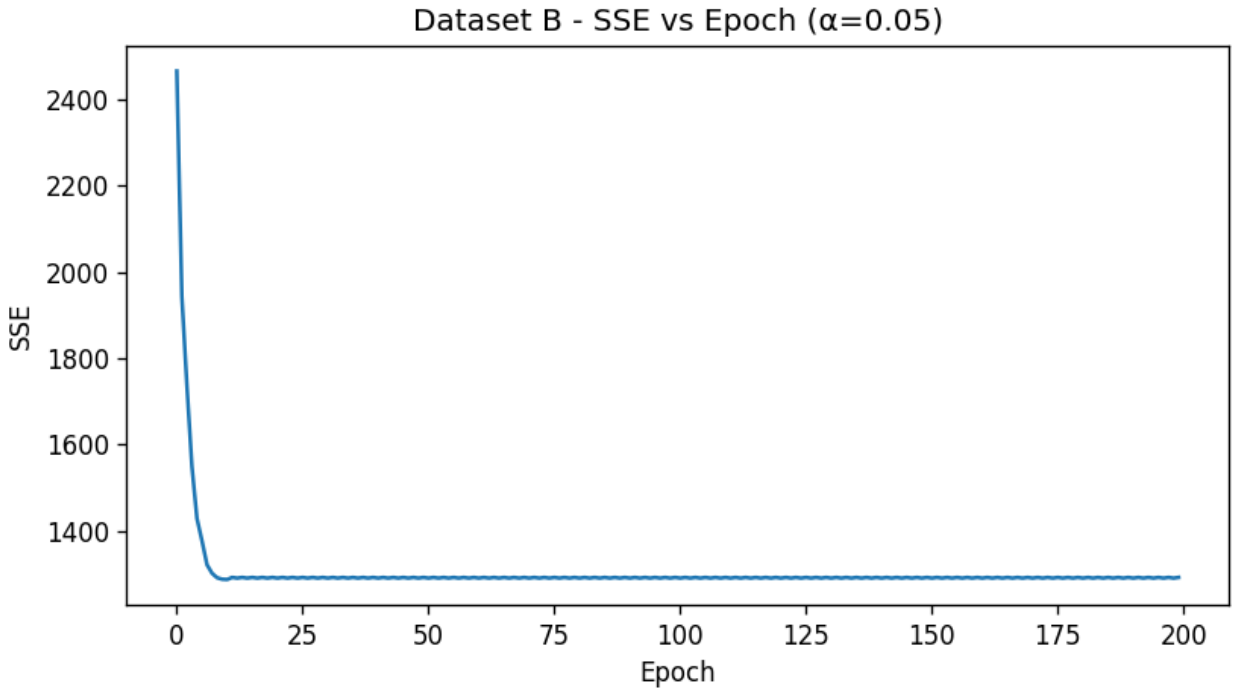
```
Final Centroids B:
 [[-2.51713824  1.4783033 ]
 [ 5.25769656  0.18060121]]
```



Dataset B - Final Prototype Clustering with Centroids



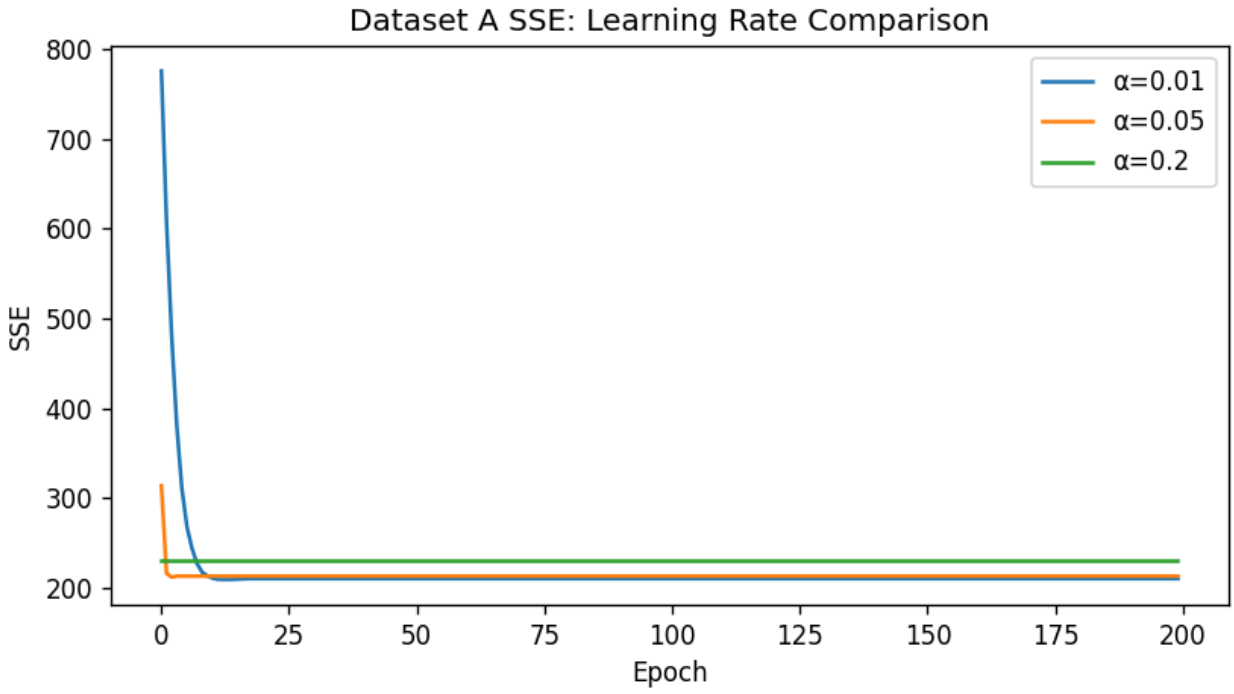
Dataset B - Centroid Positions at Epochs 50, 100, 150, 200



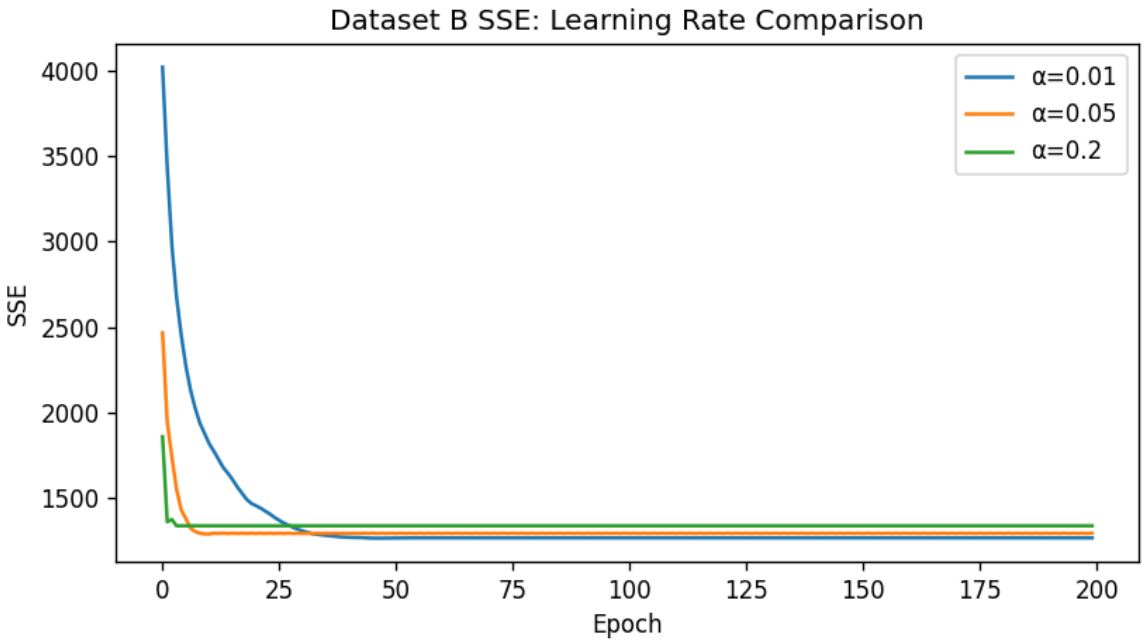
Dataset B - SSE vs Epoch (α = 0.05)

2.4 Learning Rate Comparison

The full clustering procedure was repeated for α = 0.01, α = 0.05, and α = 0.2, with the random seed reset to 172 for each run to ensure a fair comparison.



Dataset A - SSE Curves for Three Learning Rates

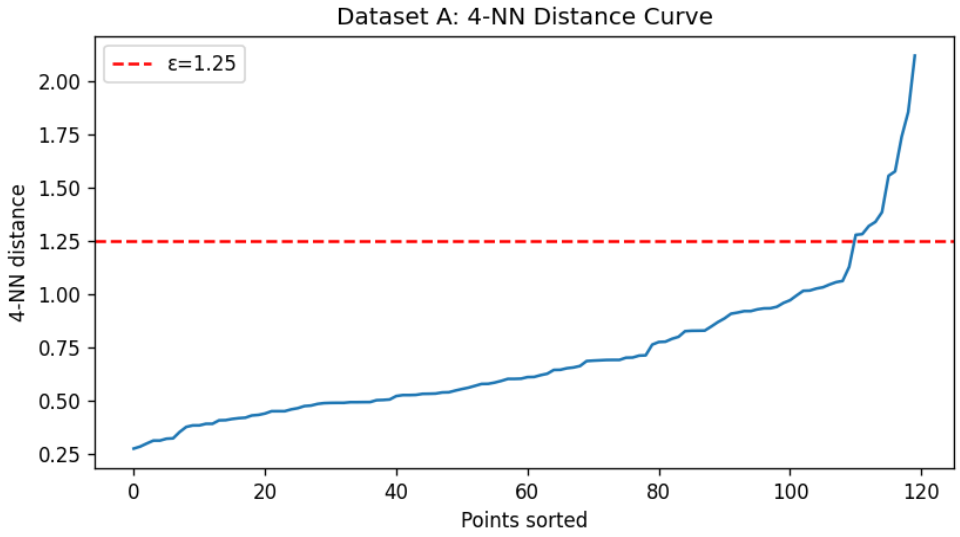


Dataset B - SSE Curves for Three Learning Rates

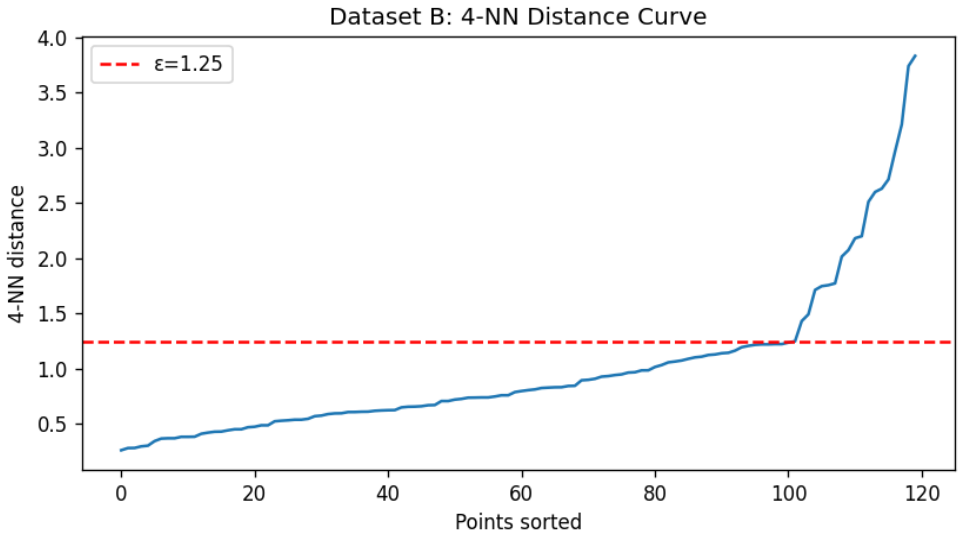
Task 3: DBSCAN

3.1 Parameter Selection via 4-NN Distance Plot

DBSCAN was implemented from scratch using Euclidean distance and minPts = 4. For each point, the distance to its fourth nearest neighbour was computed. These distances were sorted in ascending order and plotted. The elbow point where the curve bends most sharply identifies the appropriate ε value.



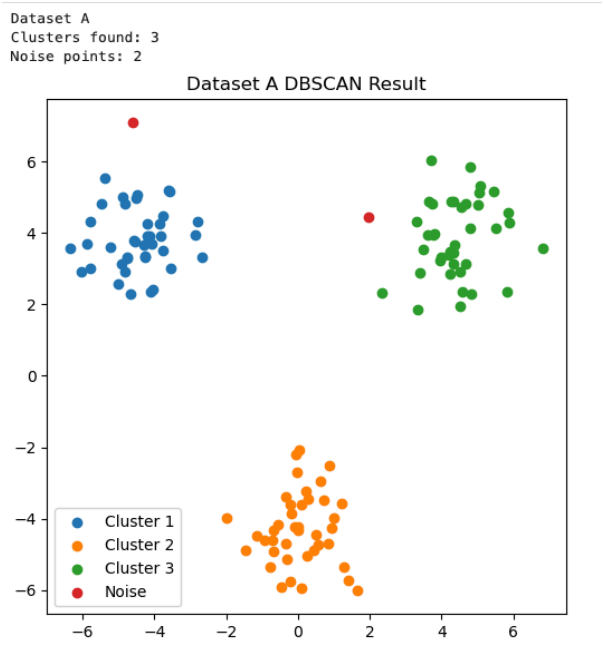
Dataset A - Sorted 4-NN Distance Curve (ε = 1.25 selected)



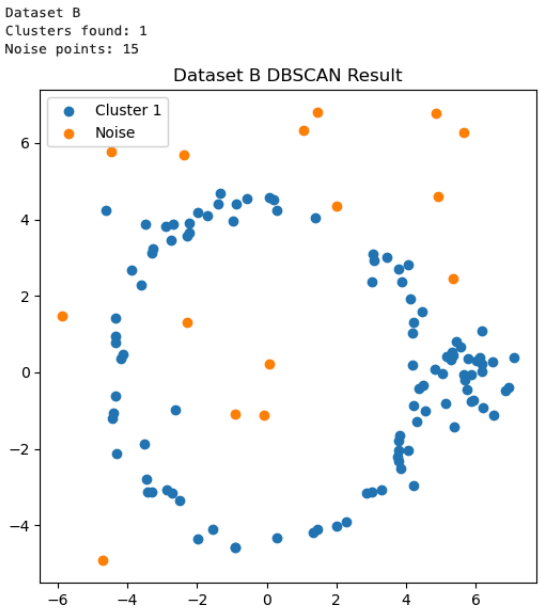
Dataset B - Sorted 4-NN Distance Curve (ε = 1.25 selected)

For both datasets the curve exhibits a clear elbow at approximately distance 1.25, which was selected as ε for both runs.

3.2 DBSCAN Results



Dataset A - DBSCAN Clustering Result



Dataset B - DBSCAN Clustering Result

DBSCAN correctly identifies 3 compact clusters in Dataset A with only 2 noise points. For Dataset B, the ring structure at radius ≈ 4.5 is not recovered as a single cluster with ε = 1.25 due to the angular gaps between ring points exceeding ε in some regions, resulting in only 1 detected cluster and 15 noise points.

Task 4: Comparison and Short Analysis

4.1 Manual Silhouette Score

The silhouette score was implemented manually. For each non-noise point i, a(i) is the mean intra-cluster distance to all other points in the same cluster, and b(i) is the minimum mean distance to points in any other cluster. The silhouette coefficient is s(i) = (b(i) − a(i)) / max(a(i), b(i)). Points in singleton clusters receive score 0. For DBSCAN, noise points (label = −1) are excluded from the calculation.

4.2 Final Comparison Table

FINAL COMPARISON TABLE

Dataset	Algorithm	Clusters	Noise	Final SSE	Silhouette
A	Prototype	3	0	213.35	0.808
A	DBSCAN	3	2	–	0.814
B	Prototype	2	0	1292.76	0.514
B	DBSCAN	1	15	–	0.0

Note: Final SSE is reported for the prototype method only (not applicable to DBSCAN). Dataset B DBSCAN silhouette = 0.000 because only 1 cluster was found; silhouette is undefined for a single cluster and is set to 0.

4.3 Short Analysis

Which dataset is more suitable for the prototype adjustment method?

Dataset A is more suitable for the manual prototype-adjustment method. It contains three well-separated, compact Gaussian clusters with identity covariance, making the clusters convex and roughly spherical. Centroid-based directional updates are most effective when clusters conform to this geometry, as the directional update rule moves each centroid incrementally toward the true cluster centre. The silhouette score of 0.808 confirms excellent separation. The SSE curve for Dataset A also shows rapid convergence within the first 50 epochs and then stabilises, indicating the algorithm settles efficiently onto the correct partition. Dataset B, by contrast, contains a ring structure that is fundamentally non-convex; a centroid placed inside the ring cannot accurately represent the ring boundary, leading to a much higher SSE of 1292.76 and a moderate silhouette of 0.514.

Which dataset is more suitable for DBSCAN?

Dataset B is, in principle, more suitable for DBSCAN because the method is designed to detect arbitrary-shaped clusters and to explicitly label low-density regions as noise. The ring, blob, and uniform noise components of Dataset B are exactly the structures DBSCAN is built for. However, the chosen $\varepsilon = 1.25$ is too small relative to the angular spacing between ring points, causing the ring to fragment. A larger ε (e.g., 1.5–2.0 for the ring) or a k-NN analysis targeted at the ring structure would recover it as a single cluster. Dataset A is also handled well by DBSCAN (3 clusters, only 2 noise points, silhouette 0.814) because the Gaussian clusters are dense and well-separated, satisfying DBSCAN's density assumptions.

How does learning rate affect convergence?

The learning rate α controls the fixed step size applied to each centroid coordinate after each point assignment. With $\alpha = 0.01$, centroids move very slowly; the SSE curve descends gradually and converges to a stable minimum only toward epoch 150–200. While the final SSE is competitive, convergence is slow, and the algorithm is susceptible to spending many epochs far from the true solution.

With $\alpha = 0.05$, the algorithm converges noticeably faster typically within the first 50–80 epochs and the SSE stabilises at a level comparable to $\alpha = 0.01$. This represents the best balance between speed and stability.

With $\alpha = 0.2$, initial descent is the fastest, but centroids overshoot the cluster centre repeatedly with each point update, causing oscillation. The SSE curve shows persistent high-frequency fluctuation rather than smooth convergence, and the final SSE is higher and less consistent across runs. This instability is more pronounced for Dataset B, where the cluster geometry is less forgiving of large centroid steps.